# Marshall Bruner

+1 719-553-9497 | brunerm99@gmail.com | marshall-bruner
brunerm99 | @MarshallBrunerRF | marshallbruner.com

*RF hardware, signal processing, and weather radar deployment | Teaching 50k+ engineers on YouTube*

## EXPERIENCE

**Rainmaker Technology Corporation** **Aug 2025 — Oct 2026**
*Head Radar Systems Engineer* *Boulder, CO*

- Led radar engineering for a cloud-seeding company: set S-, X-, and K-band sensor strategy and owned radar hardware, FPGA / embedded software, signal processing, and deployment tooling
- Designed a 24 GHz FMCW cloud-profiling radar transceiver (8-layer Rogers PCB, 450+ components, KiCad) from architecture through fab and bring-up, including a bistatic redesign for 15+ dB sensitivity gain and GaN PA bias sequencing
- Owned its software stack: SystemVerilog gateware on Zynq UltraScale+ (PLL control, LVDS ADC capture, DMA), C++ control server, Python capture and Doppler processing pipeline, and SCPI-automated bench characterization
- Operated and deployed a 25 kW X-band dual-pol scanning weather radar: scan-control and monitoring service with Slack / SNMP alerting, remote networking, and a mountain-top installation
- Architected a solid-state pulse-compression X-band dual-pol successor radar (transceiver design, PA trade study, sensitivity budget, business case) and built terrain-aware coverage-planning and radar performance-modeling tools used to site radars in 12 program regions

**CSU-CHILL National Radar Facility** **Jan 2025 — Present**
*Radar Systems Engineer (part-time)* *Greeley, CO*

- Built the facility's multi-radar web display (Flask, OpenLayers, PostgreSQL, Docker) serving real-time and historical PPI / RHI views for 8 radars and 14 polarimetric products, with client-side GPU rendering and a multi-layer cache
- Wrote the antenna positioner controller for a solid-state C-band radar in Go: reverse-implemented the pedestal's binary UDP protocol, PPI / RHI scan planner and queue, operator web UI, and CLI
- Maintained an X-band dual-pol signal processing chain in C (time series to moments to CF-Radial NetCDF to polarimetric products) and authored its operational volume coverage patterns
- Supported relocation and re-commissioning of a dual-pol S-band weather radar at Reagan Test Site, Kwajalein Atoll: bring-up procedure, calibration verification, and a live monitoring dashboard

**BAE Systems (DoD Clearance)** **Jul 2024 — Jul 2025**
*RF Engineer* *Westminster, CO*

- System design, schematic capture, and layout of a 2D active phased array weather radar
- Large-scale phased array systems analysis in Python and Matlab
- Technical consultant on weather radar proposals

**Ball Aerospace (DoD Clearance)** **Jun 2021 — Jul 2023**
*RF Engineer* *Westminster, CO*

- Designed and debugged multiple mission-critical RF boards under strict time constraints
- Simulated, designed, and tested an RF front-end PCB at extreme temperatures
- Schematic capture and layout of RF T/R systems (RF, digital, power), antenna range test equipment, and component test coupons
- Built an embedded webpage used to control and monitor phased array electronics; high-sensitivity noise and high-power testing

**Colorado State University** **Jan 2021 — May 2022**
*Graduate Research Assistant — Thesis focused on Radar* *Fort Collins, CO*

- Design, schematic, and layout of a modular X-band FMCW drone radar
- Signal processing and data visualization for Analog Devices' FMCW Phaser radar board; presented by ADI at IMS 2022 and IEEE-APS 2022
- Presented the work at the IEEE International Symposium on Phased Array Systems and Technology

**CSU-CHILL Radar Lab** **May 2020 — May 2021**
*Lab Assistant* *Greeley, CO*

- Built a server to host and display real-time and historical radar data using PostgreSQL, Flask, and Python
- Performed characterization and measurements of radar equipment

## PROJECTS

**Marshall Bruner RF — Radar / RF Education YouTube Channel** **Aug 2023 — Present**

- 53k subscribers and 1.4M+ views across 17 long-form animated videos on FMCW radar, CFAR, phased arrays, pulse compression, I/Q sampling, and dual-pol weather radar; top video 320k+ views
- Write, animate (Manim), narrate, and edit every video; built a reusable radar animation library (WSR-88D model, PPI displays, beam geometry, NEXRAD ingest) and contributed features to the ManimGX GPU engine
- Ship companion interactive notebooks (marimo to WASM) and a Python / Three.js 3D weather-radar animation studio with DEM terrain and 4/3-Earth beam refraction

**FMCW Radar Bootcamp — IEEE AESS RadarConf** **Mar 2023**

- Developed signal processing software and a presentation for an FMCW radar for the 2023 AESS Radar Bootcamp

## EDUCATION

**Colorado State University** **Jan 2021 — May 2022**
*M.S. Electrical Engineering —* ***GPA: 3.9*** *Fort Collins, CO*
*Thesis - Design, Deployment, and Cost Considerations for DARMA; A Low-Cost and Lightweight FMCW Radar*

**Colorado State University** **Aug 2017 — May 2021**
*B.S. Electrical Engineering —* ***GPA: 3.47*** *Fort Collins, CO*

## SKILLS & ABILITIES

**Radar / RF:** FMCW and pulsed radar design, pulse compression, dual-polarization polarimetry, phased arrays, link budgets, Doppler / spectral moment estimation, CFAR, receiver noise and sensitivity analysis, weather radar operations and deployment
**Languages:** Python (NumPy, SciPy, Pandas, PyTorch, FastAPI, Flask), Rust, Go, C / C++, SystemVerilog, Bash, SQL, Matlab, TypeScript
**Design / Simulation Tools:** KiCad, AWR Microwave Office, HFSS, ADS, LTSpice, Cadence, QucsStudio, Vivado
**Lab / Infrastructure:** VNA, spectrum analyzer, oscilloscope, noise figure and high-power test; Git, Linux, Docker, PostgreSQL, Tailscale, LaTeX, Manim
**Communication:** Technical presentations and design reviews (PDR / CDR), customer pre- and post-sales support, educational content reaching 50k+ engineers